MENOFYIA UNIVERSITY

SYSTEM ANALYSIS & DESIGN PROJECT

SECOND YEAR

2026-2027

جامعة المنوفية

كلية الحاسبات والمعلومات

2026-2027

Water Supply Complaint & Billing System

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Planning (Project Plan)

Project Identification

Project Name:

Water Supply Complaint & Billing System

Current Problem:

1- Citizens have to visit the company office or call to report a water complaint, and it is hard to follow up on it.
2- Bills are often based on estimated readings, which causes disputes and overcharging.
3- Water losses and leaks are reported late, so a large share of produced water is never paid for.
4- Complaint and billing records are paper-based or scattered, with no central data to analyze.

Description about project:

- The system lets a customer register a complaint (no water, low pressure, leak, wrong bill, meter fault) online or by phone, and automatically sends it to the right technical team.
- Each complaint gets a tracking number, so the customer can follow its status until it is closed.
- Duplicate complaints from the same area are merged into one incident.
- The system stores meter readings, calculates the bill from actual consumption, and shows the customer their reading history.
- Customers can receive e-bills and SMS reminders, and pay online.
- Managers get reports on complaints, leaks, response times and collection. This data can later be used for data analysis and ML (not implemented now).

System Request

System Request— Water Supply Complaint & Billing System	
Project Sponsor:	Water & Sanitation Company, Customer Service and Commercial Department (e.g. Menoufia branch)
Business Need:	1- Reduce billing errors and disputes. 2- Reduce water loss by reporting leaks faster. 3- Make complaint submission available to everyone, online. 4- Track every complaint until it is solved.
Business Requirements:	1- Online and phone complaint registration with tracking number. 2- Meter reading and bill calculation module. 3- Online payment and SMS/e-bill notifications. 4- Reports and dashboard for management.
Business Value:	1- Providing accurate information. 2- Saving time. 3- Saving cost. 4- Saving effort and water.
Special Issues or Constraints:	1- Deadline in 1 January. 2- Must integrate with existing customer and meter data. 3- Customer data must be protected.

Feasibility Analysis

Technical Feasibility

1. The system can be built as a web application using HTML, CSS, JavaScript and a back-end such as Java or Python, with a relational database.
2. Development tools (VS Code, IntelliJ, Git, Jira) are free or already available to the team.
3. It needs an internet connection and works on any browser, on Windows or Android.
4. SMS and online payment gateways can be added through standard APIs. These services already exist in Egypt's water sector.

Economic Feasibility

1- Estimated development cost is about **975,700 EGP** (including a 10% contingency).
2- Estimated operating cost is about **588,355 EGP per year**.
3- Estimated benefits are about **3,295,000 EGP per year** at full adoption (fewer disputes, less staff time, fewer field visits, savings on paper bills, better collection).
4- The 5-year analysis at a 15% discount rate gives an NPV ≈ 6.16 million EGP and a benefit-cost ratio ≈ 2.9, with payback within the first year (about 0.7 years).

Organizational Feasibility

Organizational feasibility assesses:

1. Management support: the water company is already moving to e-services (online complaints, payment, mobile apps), so management is likely to support the project.
2. Resource sufficiency: it needs customer-service staff, a technical team for field visits, and IT support. Staff need short training on the new system.
3. User acceptance: customers already use online bill payment, so a complaint-tracking portal fits existing habits.
4. Risk: resistance to changing paper procedures, which training and a phased rollout (one branch first) can reduce.

Time Estimation

	Planning	Analysis	Design	Implementation
Standard percentages	15%	20%	35%	30%
Estimated time based on planning time	Actual: 12 days	16 days	28 days	24 days